

UNIVERSITY OF ABERDEEN
Symbolic AI (CS502K)
Tutorial I – Introduction to A.I. & Intelligent Agents

1. Read chapter 1 of Russell and Norvig’s *Artificial Intelligence: A modern approach* and answer the items below in connection with the *Introduction* lecture.
 - (a) To what extent do the following computer systems embed artificial intelligence:
 - i. Supermarket bar code scanners
 - ii. Internet routing algorithms that respond dynamically to the state of the network.
 - (b) Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?
 - (c) Is AI a science, or is it engineering? Or neither or both? Justify your answer.
 - (d) “Surely computers cannot be intelligent – they can do only what their programmers tell them.” Is the latter statement true, and does it imply the former?
2. Read chapter 2 of Russell and Norvig’s *Artificial Intelligence: A modern approach* and answer the items below in connection with the *Intelligent Agents* lecture.
 - (a) For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate:
 - i. Every agent is rational in an unobservable environment.
 - ii. A perfectly rational poker-playing agent never loses.
 - (b) Give a PEAS description of the task environments below and characterize them in terms of the properties discussed in section 2.3.2 of the textbook
 - i. Shopping for used AI books on the Internet
 - ii. Bidding on an item at an auction.
 - (c) This exercise explores the differences between agent functions and agent programs.
 - i. Are there agent functions that cannot be implemented by any agent program?
 - ii. Suppose we keep the agent program fixed but speed up the machine by a factor of two. Does that change the agent function?
 - (d) Discuss possible agent programs for each of the following stochastic environments:
 - i. Murphy’s law: twenty-five percent of the time, the Suck action fails to clean the floor if it is dirty and deposits dirt onto the floor if the floor is clean. How is your agent program affected if the dirt sensor gives the wrong answer 10% of the time?
 - ii. Small children: At each time step, each clean square has a 10% chance of becoming dirty. Can you come up with a rational agent design for this case?
3. Familiarise yourself with the code base for the algorithms presented in the AIMA textbook; the code is available at <https://github.com/aimacode/aima-python>. You have two options to work with aima-python: (1) install Anaconda + Jupyter on your computer–recommended; (2) run aima-python directly online in a Binder Environment – worth trying.

- (a) Start exploring aima-python with the intro Jupyter notebook (e.g. <https://github.com/aimacode/aima-python/blob/master/intro.ipynb>). If you have a local aima-python (as recommended above) you start with your local version of the intro Jupyter notebook.
 - (b) The intro notebook has a link to another notebook called index (e.g. <https://github.com/aimacode/aima-python/blob/master/index.ipynb>). If you have a local aima-python (as recommended above) you start with your local version of the index Jupyter notebook.
4. The agents Jupyter notebook (e.g. <https://github.com/aimacode/aima-python/blob/master/agents.ipynb>) has a few examples of agents and environments – i.e. a BlindDog in a park. You will notice that the agents.ipynb imports code from agents.py. This is a common feature in aima-python where Jupyter notebooks use python libraries that actually implement the algorithms from the AIMA textbook.
- (a) Inspect the code for the abstract Agent and Environment classes which are defined in the `agents.py` library.
 - (b) Starting with a simple BlindDog agent and a Park environment, the agents.ipynb notebook has multiple implementations of BlindDog agents and their Park environments increasing the complexity of the environment. You will learn how complex environments increases the sophistication of the agents e.g. EnergeticBlindDog.
 - (c) The agents.ipynb also shows an example of the Wumpus environment. We will only learn building functionality to reason about individual actions an agent in the Wumpus world can take at any given stage of the agent’s journey in the Wumpus world. A fully functioning agent for the Wumpus environment is way more complex than building a BlindDog. Why?